

CENG 424 - Embedded Computer Systems

Homework #2

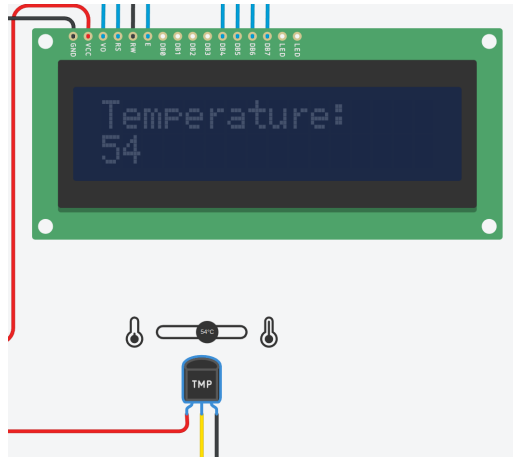
(MÜDEK ÖÇ1-PÇ4, ÖÇ1-PÇ8c)

March 23 - April 5, 2026

In this assignment, you are required to design and implement an embedded system using the Arduino platform in Tinkercad. The system should interface with a temperature sensor to measure the ambient temperature and display the measured value on an LCD screen.

Your implementation must correctly read the temperature data from the sensor, process it as necessary, and continuously update the LCD display with the current temperature value in degrees.

You are expected to build and simulate your circuit in Tinkercad and ensure that it operates reliably. The displayed information should be clear and accurate.



Notes and Restrictions

- Any form of AI-generated content, cheating, or collaboration is strictly prohibited.
- You must use a **16x2 LCD display**. The use of any other type of display is not allowed.
- You are required to use the **LiquidCrystal** library. While this library is directly available in Tinkercad, the use of any alternative libraries is not permitted.
- The LCD contrast must be properly adjusted. You may use a potentiometer to achieve an appropriate current/voltage level for clear visibility.
- The analog signal obtained from the temperature sensor must be properly scaled. You may refer to the **map** function for this purpose.
- Minor deviations ($\pm 1-2$ degrees) in the measured and displayed temperature values are acceptable.
- Your code must be written in the text mode. Do not use blocks.

Submission Guidelines

You are required to submit the following:

- A shareable Tinkercad project link containing your complete circuit design and simulation.
- A screenshot of your working system clearly showing the LCD display with the temperature value.
- Your Arduino code submitted as a **.txt** file.